Evaluation of Algorithms for Fractal Dimension Estimation in Medical Images: Complexity and Performance Analysis

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Resumo

Este estudo avalia quatro algoritmos de estimação de dimensão fractal — Box-Counting Clássico (BC), Gliding Box (GB), Differential Box-Counting (DBC) e Blanket Method (BM) — sob a perspectiva da engenharia algorítmica aplicada a imagens médicas. Limites assintóticos formais de tempo e espaço auxiliar são derivados para formulações ingênuas e otimizadas, com validação empírica em um conjunto de dados histológicos de câncer de mama. Fundamentalmente, os resultados demonstram que melhorias assintóticas teóricas nem sempre se traduzem em ganhos práticos de tempo de execução. Embora otimizações estruturais tenham proporcionado acelerações substanciais para o GB (imagens integrais) e DBC (tabelas esparsas), elas introduziram severas sobrecargas de memória ou foram neutralizadas por ineficiências de cache e condições de parada antecipada, como observado no desempenho paradoxalmente inferior do BC otimizado. A análise de efetividade revela que, enquanto BC e BM fornecem forte separação estatística entre tecidos benignos e malignos ($p < 0.01$), o poder discriminatório do GB reside na lacunaridade em vez da dimensão fractal. Por fim, este trabalho expõe o trade-off crítico entre a eficiência computacional de métodos binários, inerentemente sensíveis a limiares, e a robustez dos métodos em tons de cinza. Esses achados fornecem um arcabouço rigoroso para a seleção de algoritmos em pipelines clínicos, enfatizando que a hierarquia de memória e restrições de implementação são componentes tão críticos para o desempenho em mundo real quanto a complexidade assintótica.

Palavras-chave: Dimensão fractal; Box-counting; Complexidade algorítmica; Imagens médicas; Classificação histológica.

Abstract

This study evaluates four fractal dimension estimation algorithms—Classical Box-Counting (BC), Gliding Box (GB), Differential Box-Counting (DBC), and the Blanket Method (BM)—from an algorithmic engineering perspective applied to medical imaging. Formal asymptotic bounds for time and auxiliary space are derived across naive and structurally optimized formulations, with empirical validation on a breast cancer histology dataset. Crucially, our findings demonstrate that theoretical asymptotic improvements do not strictly translate into practical runtime gains. While structural optimizations yielded substantial speedups for GB (integral images) and DBC (sparse tables), they introduced severe memory overheads or were counteracted by cache inefficiencies and early-exit conditions, as observed in the paradoxically slower optimized BC. Effectiveness analysis reveals that while BC and BM provide strong statistical separation between benign and malignant tissues ($p < 0.01$), GB's discriminatory power relies on lacunarity rather than fractal dimension. Ultimately, this work exposes a fundamental trade-off between the computational efficiency of threshold-sensitive binary methods and the robust, albeit resource-intensive, nature of grayscale methods. These insights provide a rigorous framework for algorithm selection in clinical pipelines, emphasizing that memory hierarchy and dataset-specific implementation constraints are as critical to real-world performance as asymptotic complexity.

Keywords: Fractal dimension; Box-counting; Algorithmic complexity; Medical imaging; Histological classification.

1 Introduction

Fractal geometry provides a mathematical framework for quantifying the structural complexity and statistical self-similarity of biological phenomena that classical Euclidean geometry fails to capture (MANDELBROT, 1982). In medical image analysis, the fractal dimension (D) has been widely adopted to characterize tissue morphology, demonstrating significant clinical utility in assessing pulmonary emphysema (MISHIMA *et al.*, 1999), retinal vasculature (JELINEK; FERNANDEZ, 2000), cortical folding (FREE *et al.*, 1996), and tumour margin irregularity (RANGAYYAN; NGUYEN, 2007). Recently, fractal descriptors have evolved beyond standalone features, being integrated into hybrid machine learning pipelines to enhance model interpretability (ROBERTO *et al.*, 2021a) and guide data augmentation strategies (BORGUE *et al.*, 2025a; LOPES; BETROUNI, 2011).

Despite this proven discriminative utility, existing literature focuses heavily on diagnostic effectiveness while largely neglecting the underlying computational trade-offs. Contemporary clinical datasets routinely comprise high-resolution images where algorithmic efficiency is no longer a secondary concern, but a strict practical constraint. Consequently, the critical research question must shift from *whether* fractal dimension is discriminative to *how efficiently and robustly* it can be extracted under realistic resource limitations.

This study investigates the relationship between theoretical algorithmic complexity, implementation choices, memory utilization, and practical execution time. We evaluate four fundamental estimation algorithms that span the principal design axes of fractal image analysis (binary versus grayscale inputs, and disjoint versus dense-overlapping covering strategies): Classical Box-Counting (BC) (MANDELBROT, 1982), Gliding Box (GB) (ALLAIN; CLOITRE, 1991), Differential Box-Counting (DBC) (SARKAR; CHAUDHURI, 1994), and the Blanket Method (BM) (PELEG *et al.*, 1984).

We hypothesize that theoretical asymptotic improvements do not necessarily translate into practical runtime gains. While structural optimizations (such as integral images or sparse tables) reduce mathematical complexity, their real-world performance is heavily mediated by memory hierarchy bottlenecks, cache behavior, early-exit conditions, and preprocessing requirements. To test this hypothesis, this work addresses the following concise objectives:

- **Theoretical vs. Empirical Scaling:** Derive formal asymptotic time and auxiliary space bounds for naive and structurally optimized formulations of all four methods, explicitly contrasting theoretical predictions with observed wall-clock performance on a standard medical dataset.
- **Resource Trade-offs:** Quantify the execution and memory overheads imposed by advanced data structures and threshold-based preprocessing steps, evaluating their feasibility for clinical deployment.
- **Discriminatory Effectiveness:** Assess the statistical separation between pathological and healthy tissues generated by each algorithm, culminating in a synthesized trade-off analysis that balances computational cost against diagnostic value.

The remainder of this article is structured as follows: Section 2 establishes the theoretical foundations of the evaluated methods. Section 3 details the algorithm designs and their asymptotic complexities. Section 4 outlines the experimental methodology. Section 5 analyzes the empirical results, explicitly addressing the divergence between theory and practice. Section 6 discusses threats to validity, and Section 7 concludes with evidence-driven guidelines for algorithm selection in medical image analysis.

2 Theoretical Foundation

2.1 Fractal Analysis and Dimension Estimation

Fractal sets exhibit *statistical self-similarity*, maintaining structural complexity across observation scales (MANDELBROT, 1982). This property is characterized by power-law scaling, where a characteristic measure $M(\varepsilon)$ obtained at scale ε satisfies

$$M(\varepsilon) \sim C \cdot \varepsilon^{-D}, \tag{1}$$

with C acting as a proportionality constant and D representing the fractal dimension (FALCONER, 2003). Taking logarithms of both sides yields a linear relationship:

$$\log M(\varepsilon) = \log C - D \log \varepsilon. \tag{2}$$

D is estimated as the negative slope of a least-squares regression over the pairs $\{(\log \varepsilon_i, \log M(\varepsilon_i))\}_{i=1}^{m}$ across m discrete scales.

As the theoretical Hausdorff dimension (HAUSDORFF, 1919) is intractable on discrete grids, the Minkowski–Bouligand (box-counting) dimension D_B serves as the standard approximation (FALCONER, 2003):

$$D_B = \lim_{\varepsilon \to 0} \frac{\log N(\varepsilon)}{\log(1/\varepsilon)}, \tag{3}$$

where $N(\varepsilon)$ is the minimum number of boxes of side ε needed to cover the set. The regression's coefficient of determination (R^2) validates the presence of consistent power-law scaling, a prerequisite for interpreting D_B as a meaningful structural descriptor (LOPES; BETROUNI, 2011).

2.2 Classical Box-Counting Method

The Classical Box-Counting (BC) method discretizes Eq. (3) for binary images (MANDELBROT, 1982; LIEBOVITCH; TOTH, 1989). Given an image I of $N = W \times H$ pixels, the domain is partitioned into $\lceil W/\varepsilon \rceil \times \lceil H/\varepsilon \rceil$ disjoint cells of side ε. A cell is considered *occupied* if it contains at least one foreground pixel:

$$N(\varepsilon) = \sum_{i=0}^{\lceil W/\varepsilon \rceil - 1} \sum_{j=0}^{\lceil H/\varepsilon \rceil - 1} \mathbf{1}[\exists\, (x, y) \in B_{ij}(\varepsilon) : I(x, y) = 1]\,. \tag{4}$$

Applying BC to grayscale medical images requires prior binarization, typically via Otsu's global thresholding (OTSU, 1979). While binarization reduces the representational complexity, it irreversibly discards intensity gradients, making D_B highly sensitive to the threshold parameter t^*.

To bypass exhaustive pixel scanning, optimized implementations pre-compute the *integral image* $P[x][y] = \sum_{i \leq x, j \leq y} I[i][j]$ via the recurrence (CROW, 1984):

$$P[x][y] = I[x][y] + P[x-1][y] + P[x][y-1] - P[x-1][y-1]. \tag{5}$$

The foreground pixel count within any rectangular cell is then extracted in $\mathcal{O}(1)$ time:

$$\begin{aligned} \sigma(x_1, y_1, x_2, y_2) = P[x_2][y_2] - P[x_1-1][y_2] \\ - P[x_2][y_1-1] + P[x_1-1][y_1-1]. \end{aligned} \tag{6}$$

2.3 Gliding Box Method

The Gliding Box (GB) method (ALLAIN; CLOITRE, 1991) evaluates a dense sliding window rather than a disjoint grid. A box of side L is centered at every valid pixel position (x, y), defining the *box mass* as:

$$m_{x,y}(L) = \sum_{i=0}^{L-1} \sum_{j=0}^{L-1} I[x+i][y+j]. \tag{7}$$

Across $T(L) = (W - L + 1)(H - L + 1)$ valid placements, the *mass-frequency distribution* $n(m, L)$ dictates the normalized probability $P(m, L) = n(m, L)/T(L)$. D_B is estimated by substituting the mean box mass into Eq. (2) (PLOTNICK *et al.*, 1996):

$$\mu(L) = \sum_{m=0}^{L^2} m\, P(m, L). \tag{8}$$

GB also extracts *lacunarity*, capturing the translational heterogeneity of the spatial mass distribution:

$$\Lambda(L) = \frac{\mu^{(2)}(L) - [\mu(L)]^2}{[\mu(L)]^2}, \tag{9}$$

where $\mu^{(2)}(L) = \sum_m m^2 P(m, L)$. Homogeneous textures yield $\Lambda \approx 0$, whereas clustered geometries produce large Λ (ALLAIN; CLOITRE, 1991; TOLLE; MCJUNKIN; GORSICH, 2008). Integral images (Eq. (6)) optimize mass evaluations to $\mathcal{O}(1)$ (CROW, 1984), a critical reduction given the dense $\Theta(N)$ overlapping evaluations per scale.

2.4 Differential Box-Counting Method

The Differential Box-Counting (DBC) method (SARKAR; CHAUDHURI, 1994) avoids binarization by modeling the grayscale image $I \in \{0, \ldots, G-1\}$ as a three-dimensional topographic relief. For a cell of side ε, the intensity axis is partitioned into discrete bands of height $\delta = \lceil G/\lceil W/\varepsilon \rceil \rceil$. The number of boxes required to encompass the intensity gradient within cell (i, j) is:

$$n_{ij}(\varepsilon) = \left\lceil \frac{I_{\max}^{ij} - I_{\min}^{ij}}{\delta} \right\rceil + 1, \tag{10}$$

where $I_{\max}^{ij}$ and $I_{\min}^{ij}$ denote local extrema. The total covering count $N(\varepsilon) = \sum_{i,j} n_{ij}(\varepsilon)$ determines D_B. While DBC preserves the structural fidelity of grayscale inputs, optimized execution relies on 2D sparse tables to reduce the local extrema search from $\Theta(\varepsilon^2)$ to $\mathcal{O}(1)$ queries.

2.5 Blanket Method

The Blanket Method (BM) (PELEG *et al.*, 1984) computes D_B by quantifying the apparent area growth of the intensity surface under sequential morphological operations. The upper (u_ε) and lower (b_ε) *blankets* at dilation level ε are recursively defined:

$$u_\varepsilon(x, y) = \max\Big(u_{\varepsilon-1}(x, y) + 1,\ \max_{(s,t)\in\mathcal{N}} u_{\varepsilon-1}(s, t)\Big), \tag{11}$$

$$b_\varepsilon(x, y) = \min\Big(b_{\varepsilon-1}(x, y) - 1,\ \min_{(s,t)\in\mathcal{N}} b_{\varepsilon-1}(s, t)\Big), \tag{12}$$

initialized at $u_0 = b_0 = I$, with $\mathcal{N}$ representing the 4-connected neighborhood. The enclosed *blanket volume* is:

$$V(\varepsilon) = \sum_{x,y} \big[u_\varepsilon(x, y) - b_\varepsilon(x, y)\big]. \tag{13}$$

The apparent surface area $A(\varepsilon) = \big[V(\varepsilon) - V(\varepsilon-1)\big]/2$ substitutes $M(\varepsilon)$ in Eq. (2) to derive the fractal dimension. Operating strictly on intensity profiles, BM avoids thresholding artifacts. While a naive implementation incurs a neighborhood-dependent computational cost, separable monotonic deque filters (LEMIRE, 2006) optimize the 2D morphological extrema to $\Theta(N)$ total operations per scale.

3 Algorithm Project and Analysis

This section details the algorithmic design and formal asymptotic analysis of the four fractal dimension estimators. For each method, we first define a naive brute-force formulation, identify its computational bottleneck, and subsequently derive a structurally optimized counterpart. Throughout this analysis, $N = W \times H$ denotes the total image pixel count, $m = |\mathcal{S}|$ the number of evaluated scales, and ε (or L) the current box side length. Asymptotic bounds follow standard Bachmann–Landau notation ($\mathcal{O}$, Θ, Ω).

3.1 Classical Box-Counting (BC)

3.1.1 Naive Formulation

Algorithm 1 implements Eq. (4) through exhaustive cell scanning. For each scale ε, the grid contains $\Theta(N/\varepsilon^2)$ cells. In the worst case (a fully foreground cell), evaluating a cell requires ε^2 pixel inspections; in the best case (first pixel is foreground), it requires one. The worst-case per-scale cost is:

$$T_{\text{BC,naive}}(\varepsilon) = \Theta\left(\frac{N}{\varepsilon^2}\right) \cdot \mathcal{O}(\varepsilon^2) = \mathcal{O}(N). \tag{14}$$

The strict lower bound is $\Omega(N/\varepsilon^2)$ since every cell must be visited. Summing over all m scales, including the $\Theta(N)$ binarization pass, yields the total execution time:

$$T_{\text{BC,naive}}(N, m) = \Theta(N) + \sum_{\varepsilon\in\mathcal{S}} \mathcal{O}(N) = \mathcal{O}(mN). \tag{15}$$

Beyond the read-only input buffer, auxiliary space is $\Theta(1)$.

Algorithm 1 Naive Classical Box-Counting

Require: Binary image $I[0 \dots W-1][0 \dots H-1]$, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon, N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$
Binarize I via Otsu's threshold t^* (OTSU, 1979)
for each $\varepsilon \in \mathcal{S}$ **do**
  $count \leftarrow 0$
  for $b_x \leftarrow 0$ **to** $\lceil W/\varepsilon \rceil - 1$ **do**
    for $b_y \leftarrow 0$ **to** $\lceil H/\varepsilon \rceil - 1$ **do**
      $occupied \leftarrow$ False
      for $dx \leftarrow 0$ **to** $\varepsilon - 1$ **do**
        for $dy \leftarrow 0$ **to** $\varepsilon - 1$ **do**
          $px \leftarrow b_x \cdot \varepsilon + dx$; $py \leftarrow b_y \cdot \varepsilon + dy$
          if $px < W \wedge py < H \wedge I[px][py] = 1$ **then**
            $occupied \leftarrow$ True; **break**
          end if
        end for
      end for
      if $occupied$ **then**
        $count \leftarrow count + 1$
      end if
    end for
  end for
  yield $(\varepsilon, count)$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i, \log N(\varepsilon_i))\}$

Algorithm 2 Optimized Box-Counting (Integral Image)

Require: Binary image $I[0 \dots W-1][0 \dots H-1]$, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon, N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$
Binarize I via Otsu's threshold t^* (OTSU, 1979)
Build integral image P via Eq. (5) ▷ $\Theta(N)$ time and space (CROW, 1984)
for each $\varepsilon \in \mathcal{S}$ **do**
  $count \leftarrow 0$
  for $b_x \leftarrow 0$ **to** $\lceil W/\varepsilon \rceil - 1$ **do**
    for $b_y \leftarrow 0$ **to** $\lceil H/\varepsilon \rceil - 1$ **do**
      $x_1 \leftarrow b_x \cdot \varepsilon$; $y_1 \leftarrow b_y \cdot \varepsilon$
      $x_2 \leftarrow \min(x_1+\varepsilon-1, W-1)$; $y_2 \leftarrow \min(y_1+\varepsilon-1, H-1)$
      if $\sigma(x_1, y_1, x_2, y_2) > 0$ **then** ▷ Eq. (6), $\mathcal{O}(1)$
        $count \leftarrow count + 1$
      end if
    end for
  end for
  yield $(\varepsilon, count)$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i, \log N(\varepsilon_i))\}$

3.1.2 Optimized Formulation

The inner double loop of Algorithm 1 represents the primary computational bottleneck. Algorithm 2 eliminates this by pre-computing the integral image P (Eq. (5)). Consequently, region sums are resolved in $\mathcal{O}(1)$ via Eq. (6) (CROW, 1984), determining cell occupancy ($\sigma > 0$) without pixel scanning.

The prefix sum is computed once in $\Theta(N)$. Subsequently, traversing the $\Theta(N/\varepsilon^2)$ cells per scale costs $\mathcal{O}(1)$ each:

$$T_{\mathrm{BC,opt}}(N, m) = \Theta(N) + \sum_{\varepsilon \in \mathcal{S}} \Theta\left(\frac{N}{\varepsilon^2}\right). \tag{16}$$

For a geometric scale set (e.g., $\varepsilon \in \{2, 4, 8 \dots\}$), the sum is dominated by the smallest scale, collapsing the total time complexity to $\Theta(N)$. This strictly improves upon the $\mathcal{O}(mN)$ naive bound. The required integral image introduces a $\Theta(N)$ memory footprint.

Table 1: Asymptotic complexity of Classical Box-Counting.

Formulation	Time	Aux. Space
Naive	$\mathcal{O}(mN)$	$\Theta(1)$
Optimized	$\Theta(N)$	$\Theta(N)$

3.2 Gliding Box (GB)

3.2.1 Naive Formulation

Algorithm 3 implements the dense sliding window mass calculation defined in Eq. (7). For a fixed scale L, scanning L^2 pixels across $\Theta(N)$ valid window positions yields a per-scale cost of:

$$T_{\mathrm{GB,naive}}(L) = \Theta(N) \cdot \Theta(L^2) = \Theta(NL^2). \tag{17}$$

Summing over m scales, the total execution time is heavily penalized by larger box sizes:

$$T_{\mathrm{GB,naive}}(N, m) = \Theta\left(N \sum_{L \in \mathcal{S}} L^2\right). \tag{18}$$

The per-scale mass histogram occupies $\mathcal{O}(L_{\max}^2)$ auxiliary space, which is generally negligible.

3.2.2 Optimized Formulation

Algorithm 4 applies the integral-image optimization to the GB method. Pre-computing P (Eq. (5)) reduces each dense window summation to $\mathcal{O}(1)$. If BC and GB operate in the same pipeline, this $\Theta(N)$ data structure is entirely shared.

With $\mathcal{O}(1)$ mass evaluations, the per-scale complexity drops to the $\Theta(N)$ valid window positions:

$$T_{\mathrm{GB,opt}}(N, m) = \Theta(N) + \sum_{L \in \mathcal{S}} \Theta(N) = \Theta(mN). \tag{19}$$

This represents a $\Theta(L^2)$ acceleration per scale over the naive implementation. While BC visits $\Theta(N/\varepsilon^2)$ disjoint cells, GB must visit $\Theta(N)$ overlapping positions regardless of scale. This algorithmic overhead enables GB to extract the richer mass-frequency distributions necessary for lacunarity estimation (Λ).

Algorithm 3 Naive Gliding Box

Require: Binary image $I[0\ldots W{-}1][0\ldots H{-}1]$, scale set $\mathcal{S}$
Ensure: Pairs $\{(L,\, \mu(L),\, \Lambda(L)) : L \in \mathcal{S}\}$
Binarize I via Otsu's threshold t^* (OTSU, 1979)
for each $L \in \mathcal{S}$ **do**
  $H[0\ldots L^2] \leftarrow 0$ ▷ mass-frequency histogram
  for $x \leftarrow 0$ **to** $W - L$ **do**
    for $y \leftarrow 0$ **to** $H - L$ **do**
      $mass \leftarrow 0$
      for $dx \leftarrow 0$ **to** $L - 1$ **do**
        for $dy \leftarrow 0$ **to** $L - 1$ **do**
          $mass \leftarrow mass + I[x{+}dx][y{+}dy]$
        end for
      end for
      $H[mass] \leftarrow H[mass] + 1$
    end for
  end for
  $T \leftarrow (W{-}L{+}1)(H{-}L{+}1)$
  Compute $\mu(L)$ and $\mu^{(2)}(L)$ from H and T
  Compute $\Lambda(L)$ via Eq. (9)
  yield $(L,\, \mu(L),\, \Lambda(L))$
end for
Estimate D_B by least-squares regression on $\{(\log L_i,\, \log \mu(L_i))\}$

Algorithm 4 Optimized Gliding Box (Integral Image)

Require: Binary image $I[0\ldots W{-}1][0\ldots H{-}1]$, scale set $\mathcal{S}$, integral image P
Ensure: Pairs $\{(L,\, \mu(L),\, \Lambda(L)) : L \in \mathcal{S}\}$
for each $L \in \mathcal{S}$ **do**
  $H[0\ldots L^2] \leftarrow 0$
  for $x \leftarrow 0$ **to** $W - L$ **do**
    for $y \leftarrow 0$ **to** $H - L$ **do**
      $mass \leftarrow \sigma(x,\, y,\, x{+}L{-}1,\, y{+}L{-}1)$ ▷ Eq. (6), $\mathcal{O}(1)$
      $H[mass] \leftarrow H[mass] + 1$
    end for
  end for
  $T \leftarrow (W{-}L{+}1)(H{-}L{+}1)$
  Compute $\mu(L)$ and $\mu^{(2)}(L)$ from H and T
  Compute $\Lambda(L)$ via Eq. (9)
  yield $(L,\, \mu(L),\, \Lambda(L))$
end for
Estimate D_B by least-squares regression on $\{(\log L_i,\, \log \mu(L_i))\}$

3.3 Differential Box-Counting (DBC)

3.3.1 Naive Formulation

Algorithm 5 directly implements the 3D grid partitioning detailed in Eq. (10). Because it operates on grayscale data, no binarization is performed. Traversing the $\Theta(N/\varepsilon^2)$ cells requires an exhaustive $\Theta(\varepsilon^2)$ pixel scan per cell to determine the local intensity extrema $(I^{ij}_{\max}, I^{ij}_{\min})$.

The resulting per-scale cost is $\Theta(N/\varepsilon^2) \cdot \Theta(\varepsilon^2) = \Theta(N)$, yielding a total time complexity of:

$$T_{\text{DBC,naive}}(N, m) = \Theta(mN). \tag{20}$$

Although the asymptotic class mirrors naive BC, the hidden constant factor is strictly larger. Two intensity comparisons are required per pixel, and early-exit mechanisms are mathematically impossible since true extrema require scanning the entire cell. Auxiliary space remains $\Theta(1)$.

Table 2: Asymptotic complexity of the Gliding Box method.

Formulation	**Time**	**Aux. Space**
Naive	$\Theta\big(N\sum_L L^2\big)$	$\mathcal{O}(L^2_{\max})$
Optimized	$\Theta(mN)$	$\Theta(N)$

Table 3: BC vs. GB optimized formulations (per scale).

Method	**Positions/scale**	**Cost/position**	**Per-scale cost**
BC (opt.)	$\Theta(N/\varepsilon^2)$	$\mathcal{O}(1)$	$\Theta(N/\varepsilon^2)$
GB (opt.)	$\Theta(N)$	$\mathcal{O}(1)$	$\Theta(N)$

3.3.2 Optimized Formulation

Algorithm 6 replaces the exhaustive $\Theta(\varepsilon^2)$ scan with offline 2D sparse tables (LIU *et al.*, 2014). The tables map regional minima and maxima queries to $\mathcal{O}(1)$ execution time.

Sparse table construction demands $\Theta(N \log N)$ time and space. Post-construction, traversing the $\Theta(N/\varepsilon^2)$ cells executes in unit time per cell. The total complexity is therefore bound by the preprocessing phase:

$$T_{\text{DBC,opt}}(N, m) = \Theta(N \log N) + \sum_{\varepsilon \in \mathcal{S}} \Theta\left(\frac{N}{\varepsilon^2}\right) = \Theta(N \log N). \tag{21}$$

While computationally superior at large scales, DBC optimization introduces a severe $\Theta(N \log N)$ memory overhead, establishing a prominent time-memory tradeoff for grayscale analyses.

Table 4: Asymptotic complexity of Differential Box-Counting.

Formulation	**Time**	**Aux. Space**
Naive	$\Theta(mN)$	$\Theta(1)$
Optimized	$\Theta(N \log N)$	$\Theta(N \log N)$

3.4 Blanket Method (BM)

3.4.1 Naive Formulation

Algorithm 7 implements the sequential morphological erosion and dilation recurrences defined in Eqs. (11)–(13). Each of the $\varepsilon_{\max}$ dilation levels requires examining the 4-connected neighborhood ($\mathcal{O}(1)$ work) for all N pixels.

The total execution time evaluates to:

$$T_{\text{BM,naive}}(N, \varepsilon_{\max}) = \Theta(N \cdot \varepsilon_{\max}). \tag{22}$$

Because blanket surfaces strictly depend on their prior states, parallelizing across scales is impossible. Auxiliary space requires $\Theta(N)$ to maintain the active and previous state buffers.

Algorithm 5 Naive Differential Box-Counting

Require: Grayscale image $I[0\ldots W{-}1][0\ldots H{-}1]$, intensity range G, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon,\, N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$

for each $\varepsilon \in \mathcal{S}$ **do**
  $\delta \leftarrow \lceil G \,/\, \lceil W/\varepsilon \rceil \rceil$
  $\mathit{total} \leftarrow 0$
  for $b_x \leftarrow 0$ **to** $\lceil W/\varepsilon \rceil - 1$ **do**
    for $b_y \leftarrow 0$ **to** $\lceil H/\varepsilon \rceil - 1$ **do**
      $I_{\max} \leftarrow -\infty;\ I_{\min} \leftarrow +\infty$
      for $dx \leftarrow 0$ **to** $\varepsilon - 1$ **do**
        for $dy \leftarrow 0$ **to** $\varepsilon - 1$ **do**
          $px \leftarrow b_x{\cdot}\varepsilon{+}dx;\ py \leftarrow b_y{\cdot}\varepsilon{+}dy$
          if $px < W \,\wedge\, py < H$ **then**
            $I_{\max} \leftarrow \max(I_{\max},\, I[px][py])$
            $I_{\min} \leftarrow \min(I_{\min},\, I[px][py])$
          end if
        end for
      end for
      $\mathit{total} \leftarrow \mathit{total} + \lceil (I_{\max}{-}I_{\min})/\delta \rceil + 1$
    end for
  end for
  yield $(\varepsilon,\, \mathit{total})$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\, \log N(\varepsilon_i))\}$

Algorithm 6 Optimized Differential Box-Counting (Sparse Tables)

Require: Grayscale image $I[0\ldots W{-}1][0\ldots H{-}1]$, intensity range G, scale set $\mathcal{S}$
Ensure: Pairs $\{(\varepsilon,\, N(\varepsilon)) : \varepsilon \in \mathcal{S}\}$

Build 2D sparse table $\mathcal{T}_{\max}$ for range maximum ▷ $\Theta(N \log N)$ time and space (LIU *et al.*, 2014)
Build 2D sparse table $\mathcal{T}_{\min}$ for range minimum ▷ $\Theta(N \log N)$ time and space
for each $\varepsilon \in \mathcal{S}$ **do**
  $\delta \leftarrow \lceil G \,/\, \lceil W/\varepsilon \rceil \rceil$
  $\mathit{total} \leftarrow 0$
  for $b_x \leftarrow 0$ **to** $\lceil W/\varepsilon \rceil - 1$ **do**
    for $b_y \leftarrow 0$ **to** $\lceil H/\varepsilon \rceil - 1$ **do**
      $x_1 \leftarrow b_x{\cdot}\varepsilon;\ y_1 \leftarrow b_y{\cdot}\varepsilon$
      $x_2 \leftarrow \min(x_1{+}\varepsilon{-}1,\, W{-}1);\quad y_2 \leftarrow \min(y_1{+}\varepsilon{-}1,\, H{-}1)$
      $I_{\max} \leftarrow \mathcal{T}_{\max}.\textsc{Query}(x_1, y_1, x_2, y_2)$ ▷ $\mathcal{O}(1)$
      $I_{\min} \leftarrow \mathcal{T}_{\min}.\textsc{Query}(x_1, y_1, x_2, y_2)$ ▷ $\mathcal{O}(1)$
      $\mathit{total} \leftarrow \mathit{total} + \lceil (I_{\max}{-}I_{\min})/\delta \rceil + 1$
    end for
  end for
  yield $(\varepsilon,\, \mathit{total})$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\, \log N(\varepsilon_i))\}$

3.4.2 Optimized Formulation

Evaluating morphological extrema across an expanded $(2\varepsilon + 1) \times (2\varepsilon + 1)$ window typically incurs an $\Theta(\varepsilon^2)$ penalty per pixel. Algorithm 8 sidesteps this by exploiting the *separability* of maximum and minimum filters. A 2D extrema window decomposes into successive 1D row and column passes. Using Lemire's monotonic deques (LEMIRE, 2006), each 1D pass resolves in $\Theta(N)$ amortized time.

Algorithm 7 Naive Blanket Method

Require: Grayscale image $I[0\ldots W{-}1][0\ldots H{-}1]$, maximum dilation level $\varepsilon_{\max}$
Ensure: Pairs $\{(\varepsilon,\, A(\varepsilon)) : \varepsilon = 1, \ldots, \varepsilon_{\max}\}$

$u \leftarrow I;\ b \leftarrow I$ ▷ initialize both blankets
$V_{\text{prev}} \leftarrow \sum_{x,y} [u[x][y] - b[x][y]]$
for $\varepsilon \leftarrow 1$ **to** $\varepsilon_{\max}$ **do**
  $u' \leftarrow$ new array of size $W \times H$
  $b' \leftarrow$ new array of size $W \times H$
  for $x \leftarrow 0$ **to** $W - 1$ **do**
    for $y \leftarrow 0$ **to** $H - 1$ **do**
      $u'[x][y] \leftarrow u[x][y] + 1$
      $b'[x][y] \leftarrow b[x][y] - 1$
      for each $(s, t) \in \mathcal{N}(x, y)$ **do**
        $u'[x][y] \leftarrow \max(u'[x][y],\ u[s][t])$
        $b'[x][y] \leftarrow \min(b'[x][y],\ b[s][t])$
      end for
    end for
  end for
  $u \leftarrow u';\ b \leftarrow b'$
  $V \leftarrow \sum_{x,y} [u[x][y] - b[x][y]]$
  yield $\big(\varepsilon,\ (V - V_{\text{prev}})/2\big)$
  $V_{\text{prev}} \leftarrow V$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\, \log A(\varepsilon_i))\}$

Algorithm 8 Optimized Blanket Method (Separable Monotonic Deque)

Require: Grayscale image $I[0\ldots W{-}1][0\ldots H{-}1]$, maximum dilation level $\varepsilon_{\max}$
Ensure: Pairs $\{(\varepsilon,\, A(\varepsilon)) : \varepsilon = 1, \ldots, \varepsilon_{\max}\}$

$u \leftarrow I;\ b \leftarrow I$
$V_{\text{prev}} \leftarrow \sum_{x,y} [u[x][y] - b[x][y]]$
for $\varepsilon \leftarrow 1$ **to** $\varepsilon_{\max}$ **do**
  $u_{\text{inc}} \leftarrow u + 1$ ▷ element-wise increment, $\Theta(N)$
  $b_{\text{dec}} \leftarrow b - 1$
  $u' \leftarrow \textsc{SlidingMax2D}(u,\, K{=}3)$ ▷ separable deque, $\Theta(N)$ (LEMIRE, 2006)
  $b' \leftarrow \textsc{SlidingMin2D}(b,\, K{=}3)$ ▷ separable deque, $\Theta(N)$
  $u \leftarrow \max(u_{\text{inc}},\, u')$ ▷ element-wise, $\Theta(N)$
  $b \leftarrow \min(b_{\text{dec}},\, b')$
  $V \leftarrow \sum_{x,y} [u[x][y] - b[x][y]]$
  yield $\big(\varepsilon,\ (V - V_{\text{prev}})/2\big)$
  $V_{\text{prev}} \leftarrow V$
end for
Estimate D_B by least-squares regression on $\{(\log \varepsilon_i,\, \log A(\varepsilon_i))\}$

The total asymptotic time remains equivalent to the naive class:

$$T_{\text{BM,opt}}(N, \varepsilon_{\max}) = \Theta(N \cdot \varepsilon_{\max}). \tag{23}$$

However, the separable formulation guarantees that the computational constant remains flat regardless of the structuring element size. Table 6 consolidates the bounds across all evaluated methods.

Table 5: Asymptotic complexity of the Blanket Method.

Formulation	Time	Aux. Space
Naive	$\Theta(N\varepsilon_{\max})$	$\Theta(N)$
Optimized	$\Theta(N\varepsilon_{\max})$	$\Theta(N)$

Table 6: Consolidated asymptotic complexity of all four methods (optimized formulations).

Method	Input	Time	Aux. Space
BC (opt.)	Binary	$\Theta(N)$	$\Theta(N)$
GB (opt.)	Binary	$\Theta(mN)$	$\Theta(N)$
DBC (opt.)	Grayscale	$\Theta(N \log N)$	$\Theta(N \log N)$
BM (opt.)	Grayscale	$\Theta(N\varepsilon_{\max})$	$\Theta(N)$

4 Methodology

4.1 Materials

The experimental evaluation utilizes the UCSB Breast Cancer Histology dataset (GELASCA *et al.*, 2008), comprising 58 hematoxylin and eosin (H&E) stained images partitioned into 32 benign and 26 malignant samples. The mild class imbalance (1.23:1) does not require resampling, as the fractal dimension D is computed independently per image.

This dataset provides a rigorous baseline for algorithmic benchmarking due to its fixed resolution (896 × 768 pixels; $N = 688{,}128$), which eliminates scale-related confounding variables in wall-clock comparisons. Furthermore, the binary pathological classification establishes a concrete benchmark for assessing discriminatory effectiveness (ROBERTO *et al.*, 2021b; BORGUE *et al.*, 2025b).

All images are loaded as 8-bit grayscale arrays ($G = 256$). For binary-domain methods (BC and GB), Otsu's global thresholding (OTSU, 1979) is applied prior to fractal dimension estimation.

4.2 Experimental Environment and Protocol

Experiments were executed in a CPU-only Kaggle Notebook environment (Intel Xeon, 2 virtual cores, ≈13 GB RAM) using Python 3.10, NumPy 1.26, SciPy 1.11, and scikit-image 0.22. This sequential, single-threaded setup ensures measured times reflect baseline algorithmic complexity without GPU acceleration confounds.[1]

The evaluation scale set is fixed at $\mathcal{S} = \{2, 4, 8, 16, 32, 64\}$ ($m = 6$). This range satisfies the covering assumption, bounded by the pixel resolution and $L_{\max} = 64 \ll \sqrt{N} \approx 830$. Applying a uniform scale set across all methods ensures performance differences strictly isolate algorithmic structure rather than scale coverage variability.

[1]Source code available at: `https://github.com/gabrielmbrum`

Each algorithmic formulation was executed 10 times per image. Following a single warm-up run to bypass cold-cache effects, the subsequent 9 iterations generated the mean wall-clock time ($\bar{t}$) and standard deviation (σ_t). Runs exhibiting a coefficient of variation

$$CV = \frac{\sigma_t}{\bar{t}} \times 100\% \tag{24}$$

exceeding 5% triggered additional executions until stability was achieved, ensuring robust benchmarking (FLEMMING; WALLACE, 1986). To isolate core algorithmic performance, pre-processing times (e.g., Otsu binarization for BC and GB) were timed and reported separately.

4.3 Evaluation Metrics

Three orthogonal evaluation axes capture the distinct dimensions of the algorithmic trade-offs under study.

4.3.1 Execution Time

Computational efficiency is quantified using the mean wall-clock time ($\bar{t}$) averaged over the fixed-resolution dataset. Empirical execution times are directly contrasted with the theoretical complexity bounds derived in Section 3 to assess how constant factors, memory hierarchy constraints, and caching behaviors influence practical performance.

4.3.2 Memory Consumption

Peak auxiliary memory allocation is recorded using Python's `tracemalloc` module. The read-only input buffer ($\Theta(N)$) is excluded by measuring the differential allocation during the algorithmic core execution. This isolates the memory footprint of requisite data structures (e.g., integral images, sparse tables), permitting direct verification of the theoretical auxiliary space bounds.

4.3.3 Effectiveness

Diagnostic effectiveness is evaluated by the statistical separation between benign and malignant tissue classes. The inter-class separation is defined as:

$$\Delta D = \bar{D}_{\text{malignant}} - \bar{D}_{\text{benign}}, \tag{25}$$

where $\bar{D}_{\text{malignant}}$ and $\bar{D}_{\text{benign}}$ denote the respective class means. A larger $|\Delta D|$ indicates superior discriminatory power, aligning with the clinical premise that malignancy increases structural complexity (LOPES; BETROUNI, 2011; RANGAYYAN; NGUYEN, 2007). Statistical significance is assessed via Welch's two-sample t-test ($\alpha = 0.05$), which robustly accommodates unequal variances (WELCH, 1947).

Two complementary metrics contextualize this effectiveness. First, the coefficient of determination (R^2) from the log-log regression validates the power-law scaling assumption, a necessary condition for a valid fractal descriptor (LOPES; BETROUNI, 2011). Second, for binary methods, the robustness of D is quantified by perturbing the Otsu threshold (t^*) by $\pm 10\%$ and measuring the corresponding variation in the estimated dimension, characterizing the threshold sensitivity trade-off.

5 Results and Discussion

5.1 Pre-processing Cost

The binarization step required by the BC and GB methods, performed via Otsu's global threshold (OTSU, 1979), was timed independently on all 58 images. The mean binarization time was 1.78 ± 0.60 ms per image. This represents less than 1% of the total execution budget for either method in their optimized formulations. Consequently, this pre-processing stage is practically negligible and does not constitute a computational bottleneck for clinical deployment. This isolation confirms that the execution times reported subsequently for BC and GB accurately reflect the fundamental cost of their algorithmic cores.

5.2 Execution Time: Empirical vs. Theoretical

Table 7 summarizes the mean wall-clock execution times and empirical speedups.

Table 7: Mean wall-clock execution time per image. Speedup $= \bar{t}_{\text{naive}}/\bar{t}_{\text{opt}}$.

Method	Naive (ms)	Optimised (ms)	Speedup
BC	186.0 ± 51.0	248.4 ± 3.4	0.75×
GB	2718.5 ± 33.0	45.8 ± 2.2	59.3×
DBC	1027.1 ± 19.2	119.5 ± 43.0	8.6×
BM	409.7 ± 23.6	306.8 ± 20.1	1.3×

The empirical execution times yield a central scientific finding of this study: *asymptotic superiority alone is insufficient to predict practical performance.* This is starkly evidenced by the BC method, where the structurally optimized formulation (248.4 ms) executed slower than the naive approach (186.0 ms), yielding a paradoxical speedup of 0.75×. Although the integral image mathematically reduces the $\mathcal{O}(N)$ worst-case per-scale traversal to a strict $\Theta(N/\varepsilon^2)$, practical performance is dictated by memory hierarchy effects and implementation-level characteristics. The integral image introduces a ≈ 2.6 MB `int32` array that exceeds the L2 cache of the evaluation environment, inducing substantial memory-access latency. Conversely, the naive formulation accesses the original binary array in a strictly sequential pattern, maximizing hardware prefetcher efficiency. Furthermore, histological images possess dense foreground regions, allowing the naive inner loop to trigger early-exit conditions almost immediately, reducing the effective constant factor well below theoretical worst-case predictions.

In contrast, theoretical improvements aligned with massive practical gains for the GB method, achieving a 59.3× speedup (45.8 ms vs 2718.5 ms). Because the naive GB evaluates $\Theta(N)$ overlapping windows per scale without early-exits, the $\Theta(L^2)$ pixel additions strictly manifest in practice (costing $\approx 2.8 \times 10^9$ operations at $L = 64$). The integral image optimization completely eliminates this $\Theta(L^2)$ bottleneck, proving indispensable for the clinical deployment of GB at high resolutions.

The optimized DBC achieved an 8.6× speedup (119.5 ms vs 1027.1 ms) by utilizing sparse tables to eliminate the exhaustive $\Theta(\varepsilon^2)$ extrema search. Since true extrema computation permits no early-exit, removing the scan yields immediate wall-clock benefits. However, the high standard deviation (± 43.0 ms) stems directly from operating-system memory jitter during the massive $\Theta(N \log N)$ sparse table allocation.

Finally, the BM formulation yielded a modest 1.3× speedup (306.8 ms vs 409.7 ms). As derived theoretically, both formulations share a $\Theta(N\varepsilon_{\max})$ complexity. The minor gain arises from vectorized NumPy filtering, which minimizes Python interpreter overhead compared to explicit neighborhood loops. Substantial speedups for BM are only anticipated when expanding beyond the 3×3 morphological structuring element.

5.3 Memory Consumption

Table 8: Mean peak auxiliary memory per image.

Method	Naive (KB)	Optimised (KB)
BC	0.4	5,376.6
GB	0.3	5,376.6
DBC	0.4	413,954.6
BM	1.4	1.3

Table 8 highlights memory as a first-class design consideration. While naive formulations maintain $\mathcal{O}(1)$ or minimal $\Theta(N)$ auxiliary space (operating effectively in-place), optimizations impose significant memory-performance trade-offs.

The BC and GB optimized variants require ≈ 5.25 MB to store the shared integral image, a negligible overhead for modern desktop systems. However, the DBC optimized formulation introduces an exceptionally severe ≈ 404 MB memory footprint to maintain the 2D sparse tables. While theoretically constrained to $\Theta(N \log N)$, the absolute magnitude of this requirement has profound practical implications. Allocating nearly half a gigabyte of auxiliary memory for a single 896×768 image poses strict deployment limitations for embedded diagnostic systems, parallelized high-throughput clinical cloud environments, or application to gigapixel Whole-Slide Images (WSIs). In such environments, practitioners may be forced to revert to the slower naive DBC implementation to satisfy memory constraints.

The BM approach proves highly advantageous in this regard, operating efficiently with minimal $\Theta(N)$ auxiliary arrays (≈ 1.3 KB peak overhead), making it the most memory-efficient grayscale algorithm evaluated.

5.4 Effectiveness: Fractal Dimension and Discriminatory Power

Table 9 reports the statistical separation capabilities of the estimators. Under the evaluated dataset and experi-

Table 9: Mean fractal dimension $\bar{D}$ per class, inter-class separation ΔD, mean R^2, and Welch t-test result ($\alpha = 0.05$). Naive and optimized formulations produced identical D values.

Method	$\bar{D}_{\text{ben}}$	$\bar{D}_{\text{mal}}$	ΔD	R^2	p-value	Sig.
BC	1.877 ± 0.052	1.790 ± 0.086	-0.087	0.999	0.0001	✓
GB	-1.995 ± 0.005	-1.992 ± 0.009	$+0.003$	1.000	0.149	×
DBC	2.112 ± 0.038	2.135 ± 0.036	$+0.024$	0.992	0.020	✓
BM	0.535 ± 0.087	0.609 ± 0.070	$+0.074$	0.987	0.001	✓

mental conditions, BC and BM exhibit the most robust discriminatory power. BC yields the largest absolute separation ($|\Delta D| = 0.087$, $p = 0.0001$). Malignant tissues manifest lower fractal dimensions in the binary domain, reflecting highly fragmented, less space-filling structural patterns compared to the well-organized glandular formations of benign tissues. Similarly, BM accurately captures the heightened structural irregularity of the malignant grayscale intensity surface, yielding strongly significant separation ($p = 0.001$).

Conversely, the GB method fails to discriminate classes using the fractal dimension ($p = 0.149$). Many readers may interpret the negative $\bar{D}$ values (≈ -1.99) as computational errors; however, these strictly result from the adopted formulation and regression convention. Because GB fits the log-log regression to the *mean box mass* $\mu(L)$—which scales proportionally to L^2 in uniform regions—rather than a traditional covering count, the slope converges to $+2$. Consequently, the fractal dimension defined as $-$slope converges to -2. The failure of GB to discriminate via D demonstrates that mean box mass carries limited structural signal. Instead, its discriminatory utility relies entirely on lacunarity, which exhibited strong inter-class separation ($\bar{\Lambda}_{\text{ben}} = 0.421$ vs $\bar{\Lambda}_{\text{mal}} = 0.832$). This confirms that the spatial heterogeneity (clustering) of the tumor cells, rather than their mean mass, drives histological classification.

Finally, while BC provides strong statistical separation, a robustness analysis revealed a mean D shift of 0.085 when perturbing the Otsu threshold by $\pm 10\%$. Because this variation is essentially equivalent to the inter-class separation ($|\Delta D| = 0.087$), improper pre-processing can entirely neutralize the discriminatory signal, highlighting a severe vulnerability of binary algorithms.

5.5 Trade-off Synthesis

Table 10: Consolidated trade-off summary across all optimized formulations. ✓ = best in class; ∼ = acceptable; × = weakest.

Criterion	BC	GB	DBC	BM
Execution time	✓	✓	∼	∼
Auxiliary memory	∼	∼	×	✓
R^2 quality	✓	✓	∼	∼
Discriminatory power ($\lvert\Delta D\rvert$)	✓	×	∼	✓
Threshold robustness	×	×	✓	✓

Table 10 synthesizes the fundamental trade-offs guiding algorithm selection. The binary-versus-grayscale dichotomy is paramount: binary methods (BC, GB) execute rapidly but suffer from extreme threshold sensitivity, potentially destroying their discriminatory validity if staining protocols vary. Grayscale methods (DBC, BM) bypass binarization, guaranteeing superior reproducibility across clinical cohorts at the expense of higher computational complexity and memory utilization.

Practically, if strict image standardization is guaranteed and speed is critical, the naive BC is highly recommended. If threshold robustness is required and memory resources are constrained, BM provides strong, statistically significant discrimination without the crippling memory overhead of DBC. The optimized GB should be strictly reserved for pipelines requiring lacunarity evaluation.

6 Conclusion

This study demonstrates a fundamental reality of algorithm engineering in medical image analysis: asymptotic complexity alone is insufficient to predict practical performance. While structural optimizations—integral images, sparse tables, and separable monotonic deques—successfully reduced theoretical time bounds, their real-world deployment exposed critical bottlenecks dictated by memory hierarchy, cache utilization, and data-dependent early-exit conditions.

The empirical evidence underscores severe time-memory trade-offs across the evaluated estimators. Optimizing the Gliding Box (GB) method proved indispensable, yielding a 59.3× speedup that transformed a computationally intractable formulation into a clinically deployable solution. Similarly, the Differential Box-Counting (DBC) method achieved an 8.6× acceleration, but introduced a massive ≈404 MB memory penalty that limits its applicability in constrained environments. In stark contrast, the structurally optimized Classical Box-Counting (BC) method was paradoxically slower (0.75×) than its naive counterpart, demonstrating that the overhead of large auxiliary data structures can easily eclipse theoretical gains when cache misses and the loss of early-exit conditions dominate execution. The Blanket Method (BM) maintained an exceptionally low memory footprint but yielded only a marginal 1.3× speedup under standard morphological configurations.

Beyond computational efficiency, the findings highlight a critical dichotomy between binary and grayscale methods regarding discriminatory effectiveness. BC and BM provided the strongest statistical separation between benign and malignant histological tissues ($p < 0.01$). However, binary algorithms (BC, GB) exhibited severe vulnerability to pre-processing artifacts; a mere $\pm 10\%$ perturbation in the binarization threshold was sufficient to completely neutralize the diagnostic signal of BC. Grayscale methods (DBC, BM) bypass this vulnerability entirely, trading higher computational demand for structural robustness. Furthermore, the analysis revealed that GB cannot reliably discriminate tissues using the fractal dimension; its diagnostic utility relies exclusively on lacunarity, capturing spatial heterogeneity rather than mean structural density.

Based on these consolidated findings, we provide the following practical recommendations for algorithm selection in medical image pipelines:

- **Classical Box-Counting (BC):** Recommended strictly when rapid execution is critical and image binarization protocols are perfectly standardized to prevent threshold-induced signal degradation. The naive formulation is preferred for standard-resolution images to maximize cache efficiency.
- **Blanket Method (BM):** The optimal choice when threshold robustness and strong discriminatory power are required under strict memory constraints.
- **Differential Box-Counting (DBC):** A robust grayscale alternative that avoids binarization artifacts, provided the deployment environment can comfortably accommodate massive $\Theta(N \log N)$ memory allocations.
- **Gliding Box (GB):** Should be explicitly reserved for analytical pipelines that require the quantification of spatial clustering and heterogeneity via lacunarity, as its fractal dimension estimate lacks discriminatory power.

Ultimately, the design of future medical image analysis pipelines must treat hardware constraints—particularly peak memory allocation and cache behavior—as primary design variables alongside diagnostic power. Future investigations should evaluate these implementations across higher-resolution modalities (e.g., gigapixel Whole-Slide Images) and parallelized architectures (GPU/SIMD) to further stress-test the time-memory-accuracy boundaries established in this work.

References

ALLAIN, C.; CLOITRE, M. Characterizing the lacunarity of random and deterministic fractal sets. **Physical Review A**, v. 44, n. 6, p. 3552–3558, 1991.

BORGUE, V. A. T.; GARCIA, B. L. d. O.; JUNIOR, S. A. P.; ROBERTO, G. F.; ROZENDO, G. B.; NEVES, L. A.; MARTINS, A. S.; TOSTA, T. A. A.; NASCIMENTO, M. Z. d. Association of fractal geometry and data augmentation through gans and xai for classification of histology images. In: **Proceedings**. [S.l.: s.n.], 2025.

BORGUE, V. A. T.; GARCIA, B. L. d. O.; JÚNIOR, S. A. P.; ROBERTO, G. F.; ROZENDO, G. B.; NEVES, L. A.; MARTINS, A. S.; TOSTA, T. A. A.; NASCIMENTO, M. Z. do. Association of fractal geometry and data augmentation through GANs and XAI for classification of histology images. In: **Proceedings of the 27th International Conference on Enterprise Information Systems (ICEIS 2025)**. [S.l.: s.n.], 2025. v. 1, p. 756–762.

CROW, F. C. Summed-area tables for texture mapping. In: **Proceedings of the 11th Annual Conference on Computer Graphics and Interactive Techniques (SIGGRAPH)**. [S.l.: s.n.], 1984. p. 207–212.

FALCONER, K. **Fractal Geometry: Mathematical Foundations and Applications**. 2. ed. Chichester: John Wiley & Sons, 2003. ISBN 978-0-470-84862-3.

FLEMMING, P. J.; WALLACE, J. J. How not to lie with statistics: The correct way to summarize benchmark results. **Communications of the ACM**, v. 29, n. 3, p. 218–221, 1986.

FREE, S. L.; SISODIYA, S. M.; COOK, M. J.; FISH, D. R.; SHORVON, S. D. Three-dimensional fractal analysis of the white matter surface from magnetic resonance images of the human brain. **Cerebral Cortex**, v. 6, n. 6, p. 830–836, 1996.

GELASCA, E. D.; BYUN, J.; OBARA, B.; MANJUNATH, B. S. Evaluation and benchmark for biological image segmentation. 2008.

HAUSDORFF, F. Dimension und äußeres maß. **Mathematische Annalen**, v. 79, p. 157–179, 1919. Disponível em: ⟨http://eudml.org/doc/158784⟩.

JELINEK, H. F.; FERNANDEZ, E. Neurons and fractals: How reliable and useful are calculations of fractal dimensions? **Journal of Neuroscience Methods**, v. 81, n. 1–2, p. 9–18, 2000.

LEMIRE, D. Streaming maximum-minimum filter using no more than three comparisons per element. **Nordic Journal of Computing**, v. 13, n. 4, p. 328–339, 2006. Disponível em: ⟨https://arxiv.org/abs/cs/0610046⟩.

LIEBOVITCH, L. S.; TOTH, T. A fast algorithm to determine fractal dimensions by box counting. **Physics Letters A**, v. 141, n. 8–9, p. 386–390, 1989.

LIU, Y.; CHEN, L.; WANG, H.; JIANG, L.; ZHANG, Y.; ZHAO, J.; WANG, D.; ZHAO, Y.; SONG, Y. An improved differential box-counting method to estimate fractal dimensions of gray-level images. **Journal of Visual Communication and Image Representation**, v. 25, n. 5, p. 1102–1111, 2014.

LOPES, R.; BETROUNI, N. Fractal and multifractal analysis: A review. **Medical Image Analysis**, v. 15, n. 4, p. 634–649, 2011.

MANDELBROT, B. B. **The Fractal Geometry of Nature**. New York: W. H. Freeman, 1982.

MISHIMA, M.; HIRAI, T.; ITOH, H.; NAKANO, Y.; SAKAI, H.; MURO, S.; NISHIMURA, K.; OKU, Y.; CHIN, K.; OHI, M.; NAKAMURA, T.; BATES, J. H. T.; ALENCAR, A. M.; SUKI, B. Complexity of terminal airspace geometry assessed by fractal analysis. **Journal of Applied Physiology**, v. 87, n. 6, p. 2110–2117, 1999.

OTSU, N. A threshold selection method from gray-level histograms. **IEEE Transactions on Systems, Man, and Cybernetics**, v. 9, n. 1, p. 62–66, 1979.

PELEG, S.; NAOR, J.; HARTLEY, R.; AVNIR, D. Multiple resolution texture analysis and classification. **IEEE Transactions on Pattern Analysis and Machine Intelligence**, v. 6, n. 4, p. 518–523, 1984.

PLOTNICK, R. E.; GARDNER, R. H.; HARGROVE, W. W.; PRESTEGAARD, K.; PERLMUTTER, M. Lacunarity analysis: A general technique for the analysis of spatial patterns. **Physical Review E**, v. 53, n. 5, p. 5461–5468, 1996.

RANGAYYAN, R. M.; NGUYEN, T. M. Fractal analysis of contours of breast masses in mammograms. **Journal of Digital Imaging**, v. 20, n. 3, p. 223–237, 2007.

ROBERTO, G. F.; LUMINI, A.; NEVES, L. A.; NASCIMENTO, M. Z. do. Fractal neural network: A new ensemble of fractal geometry and convolutional neural networks for the classification of histology images. **Expert Systems with Applications**, Elsevier, v. 166, p. 114103, 2021.

ROBERTO, G. F.; LUMINI, A.; NEVES, L. A.; NASCIMENTO, M. Z. do. Fractal neural network: A new ensemble of fractal geometry and convolutional neural networks for the classification of histology images. **Expert Systems with Applications**, v. 166, p. 114103, 2021.

SARKAR, N.; CHAUDHURI, B. B. An efficient differential box-counting approach to compute fractal dimension of image. **IEEE Transactions on Systems, Man, and Cybernetics**, v. 24, n. 1, p. 115–120, 1994.

TOLLE, C. R.; MCJUNKIN, T. R.; GORSICH, D. J. An efficient implementation of the gliding box lacunarity algorithm. **Physica D: Nonlinear Phenomena**, v. 237, n. 3, p. 306–315, 2008.

WELCH, B. L. The generalization of "student's" problem when several different population variances are involved. **Biometrika**, v. 34, n. 1–2, p. 28–35, 1947.